Life Sciences und
Facility Management

DAF Onboarding

Hardware, System and Software requirements

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2025R2 | Data Analysis Fundamentals

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PC

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System and hardware requirements for PCs

Operating system

- Windows 10 (all editions)
- Windows 11

Processor (CPU) and Memory (RAM)

- 64-bit architecture
- At least 8 GB RAM (16 GB recommended for larger datasets)
- .NET Framework 4.5+ (usually pre-installed on Windows 10/11)

Software installation for PCs – part 1

✅ **Step 1: Install Python**

Python is the programming language used for data analysis.

Go to https://www.python.org/downloads

Click the yellow **“Download Python”** button.

Run the installer.

Important: Check the box that says **“Add Python to PATH”** before clicking **Install Now**.

After installation, open **Command Prompt** and type:

```
python –-version
```

You should see something like Python 3.x.x.

✅ **Step 2: Install Git**

Git helps you save your code and track changes. It works with GitHub.

Go to https://git-scm.com

Click **“Download for Windows”**.

Run the installer and choose default options.

After installation, open **Command Prompt** and type:

```
git --version
```

You should see something like git version 2.x.x.

Set your name and email:

```
git config --global user.name "Your Name"
git config --global user.email "your.email@example.com"
```

Software installation for PCs – part 2

✅ Step 3: Install VS Code

VS Code is the program where you'll write and run your Python code.

Go to https://code.visualstudio.com

Click **"Download for Windows"**.

Run the installer and choose default options.

Open VS Code after installation.

✅ Step 4: Install VS Code Extensions

Extensions add features to VS Code.

In VS Code, click the **Extensions icon** on the left (looks like 4 squares).

Search Search and install these:

- **Python**
- **Jupyter**
- **GitHub Pull Requests and Issues**

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Mac

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System and hardware requirements for Mac computers

Operating system

- macOS 11 Big Sur
- macOS 12 Monterey
- macOS 13 Ventura
- macOS 14 Sonoma

Processor (CPU) and Memory (RAM)

- Intel Core i5 or Apple M1/M2 chip
- Dual-core minimum; quad-core recommended for better performance with large datasets
- Minimum: 8 GB (16 GB recommended for larger datasets)

Software installation for Macs – part 1

✅ Step 1: Install Homebrew

Homebrew is a tool that helps you install software easily.

Open the **Terminal** (find it in Applications > Utilities).

Copy and paste this command, then press Enter:

```
/bin/bash -c "$(curl -fsSL
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

Wait for it to finish. It may ask for your password.

✅ Step 2: Install Git

Git helps you save your code and track changes. It works with GitHub.

In the Terminal, type:

```
brew install git
```

After it installs, set your name and email:

```
git config --global user.name "Your Name"
git config --global user.email "your.email@example.com"
```

Software installation for Macs – part 2

✅ **Step 3: Install Python**

Python is the programming language you'll use for data analysis.

In the Terminal, type:

```
brew install python
```

After it installs, check the version:

```
python3 –version
```

✅ **Step 4: Install VS Code**

VS Code is the program where you'll write and run your Python code.

Go to https://code.visualstudio.com

Click **"Download for macOS"**.

Open the downloaded file and drag **Visual Studio Code** into your **Applications** folder.

Open VS Code.

✅ **Step 5: Install VS Code Extensions**

Extensions add features to VS Code.

In VS Code, click the **Extensions icon** on the left (looks like 4 squares).

Search Search and install these:

- **Python**
- **Jupyter**
- **GitHub Pull Requests and Issues**

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Chromebook

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System and hardware requirements for Chromebooks

Your Chromebook must support Linux (Beta), also known as Crostini. This feature is available on ChromeOS 80 and later (Most Chromebooks released in 2019 or later)

Linux (Beta) must be enabled to install programming tools like Python and VS Code.

Processor (CPU) and Memory (RAM)

- Processor: Intel Celeron or better (ARM processors may have limitations)
- Minimum: 8 GB (16 GB recommended for larger datasets)
- Storage: 64 GB eMMC or SSD (more if working with large files)

Software installation for Chromebooks – part 1

✅ Step 1: Turn on Linux (Beta)

Linux lets you run programming tools on your Chromebook.

Click the **clock** in the bottom-right corner.

Click the **Settings gear** ⚙️ .

Scroll down and click **“Advanced” > “Developers”**.

Find **Linux development environment** and click **“Turn On”**.

Follow the setup steps (it may take a few minutes).

✅ Step 2: Install Git

Git helps you save your code and track changes. It’s used with GitHub.

Open the **Terminal** (you’ll find it in your app launcher).

Type this command and press Enter:

In a Shell terminal

```
sudo apt install git
```

After it installs, set your name and email:

In a Shell terminal

```
git config --global user.name "Your Name"
git config --global user.email "your.email@example.com"
```

Software installation for Chromebooks – part 2

✅ **Step 3: Install Python**

Python is the programming language you'll use for data analysis.

In the Terminal, type:

```
sudo apt install python3 python3-pip
```

After it installs, check the version:

```
python3 –-version
```

✅ **Step 4: Install VS Code**

VS Code is the program where you'll write and run your Python code.

- Go to https://code.visualstudio.com
- Download the **.deb file** (for Debian/Ubuntu).
- Open the **Files app**, find the downloaded file.
- Right-click it and choose **"Install with Linux (Beta)"**.

✅ **Step 5: Install VS Code Extensions**

Extensions add features to VS Code.

In VS Code, click the **Extensions icon** on the left (looks like 4 squares).

Search Search and install these:

- **Python**
- **Jupyter**
- **GitHub Pull Requests and Issues**

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Alternative Setup

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Alternative setup

In case your laptop does not meet the minimum hardware and software requirements, you can also complete the module using a hosted python development environment such as Google Colab.

What is Google Colab?

Google Colab (short for Collaboratory) is a free tool from Google that lets you write and run Python code in your browser. It’s like Jupyter Notebook, but online.

What You Need

A Google account (e.g., Gmail)

A modern web browser (like Chrome or Firefox)

Internet access

Step-by-Step Instructions

Go to https://colab.research.google.com

Sign in with your Google account.

Click “New Notebook” to start coding.

Internet access & Software installation rights

For the data analysis module, we will make extensive use of the internet. We will be using zoom for screen sharing and presentations during class. Depending on the topics covered in class, we will be accessing many web sites (github.com, data.world, kaggle.com, www.ebi.ac.uk (genomics, molecular biology, …), openneuro.org (MRI, fMRI, EEG, …), clinicaltrials.gov etc.
It is critical to ensure that the laptop you are using for this module is not restricted by a firewall blocking access to these websites.

To install the required software, you also need to ensure that you have the appropriate access rights to your laptop (some company laptops do not allow the installation of additional software by employees).

Contact

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